

Concurrency Benchmark Results

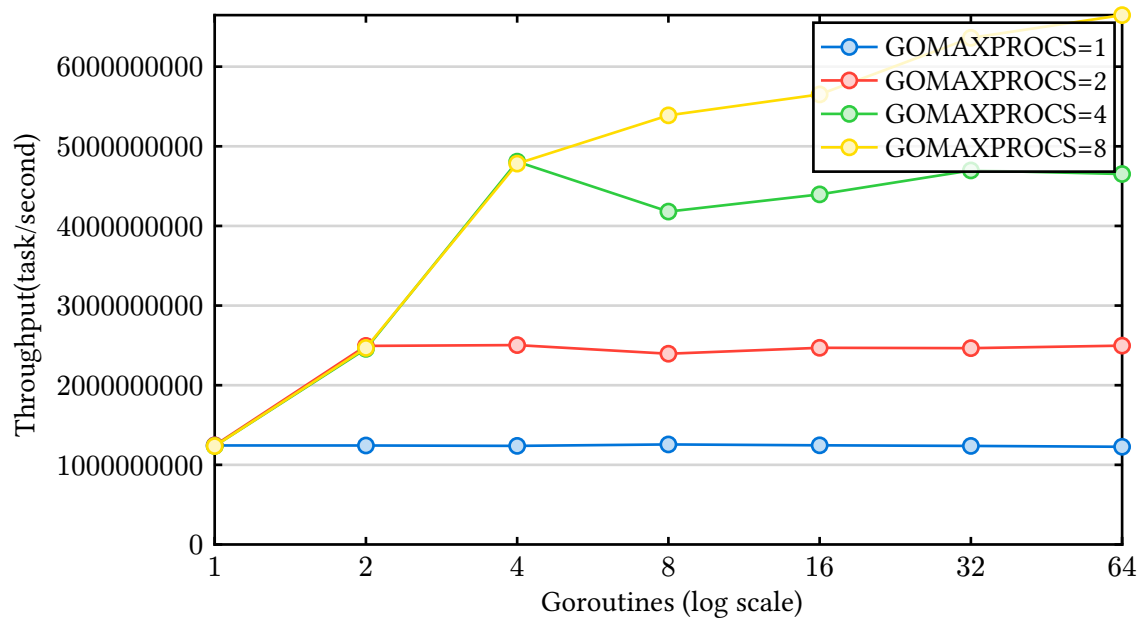


Figure 1: CPU-bound Throughput Comparison

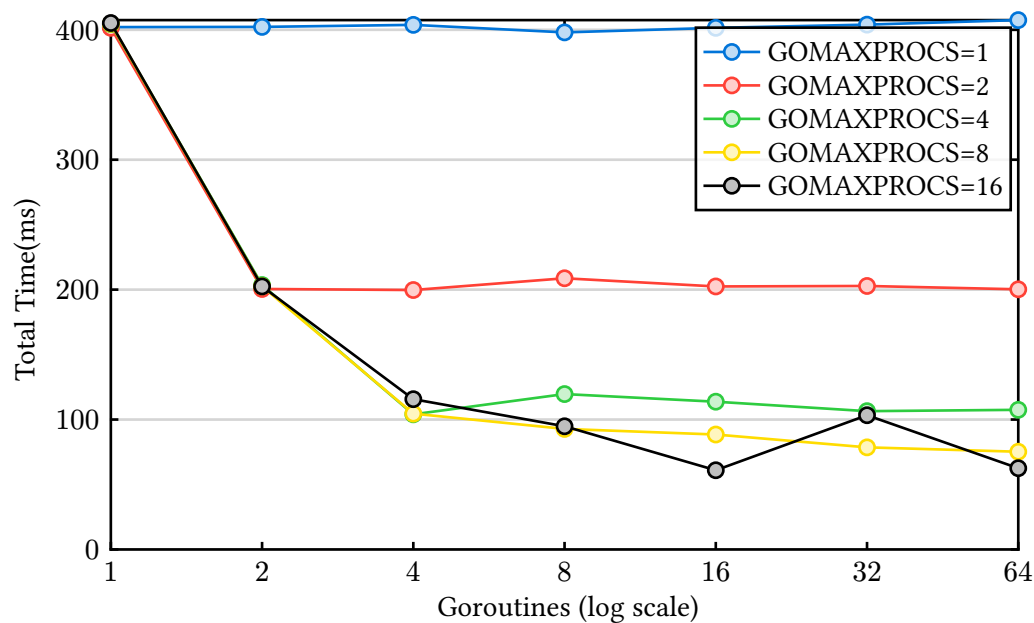


Figure 2: CPU-bound Throughput Comparison

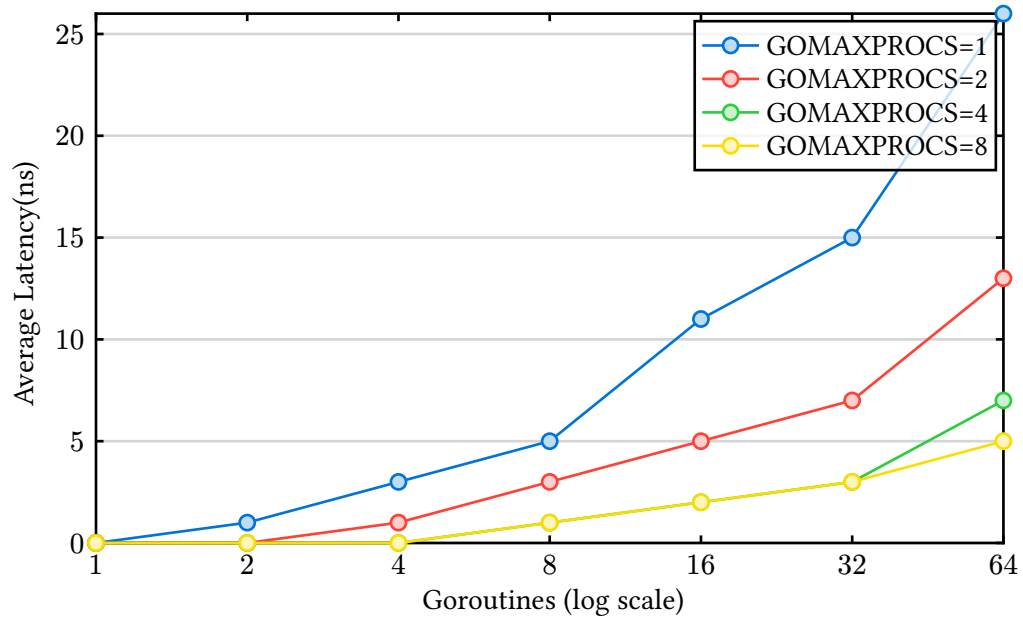


Figure 3: CPU-bound Throughput Comparison

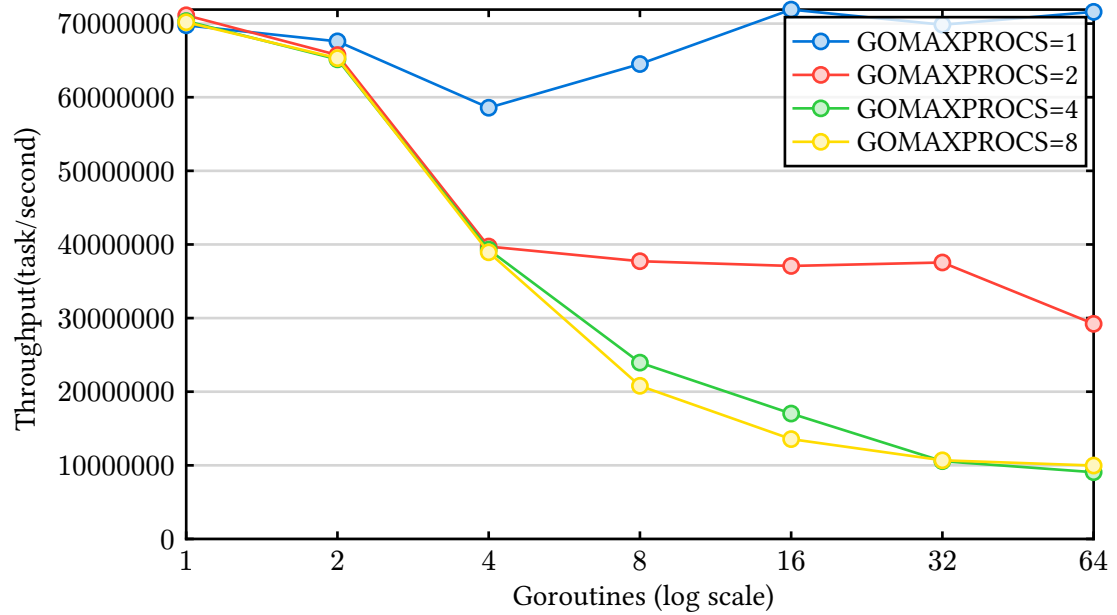


Figure 4: Mutex-bound Throughput Comparison

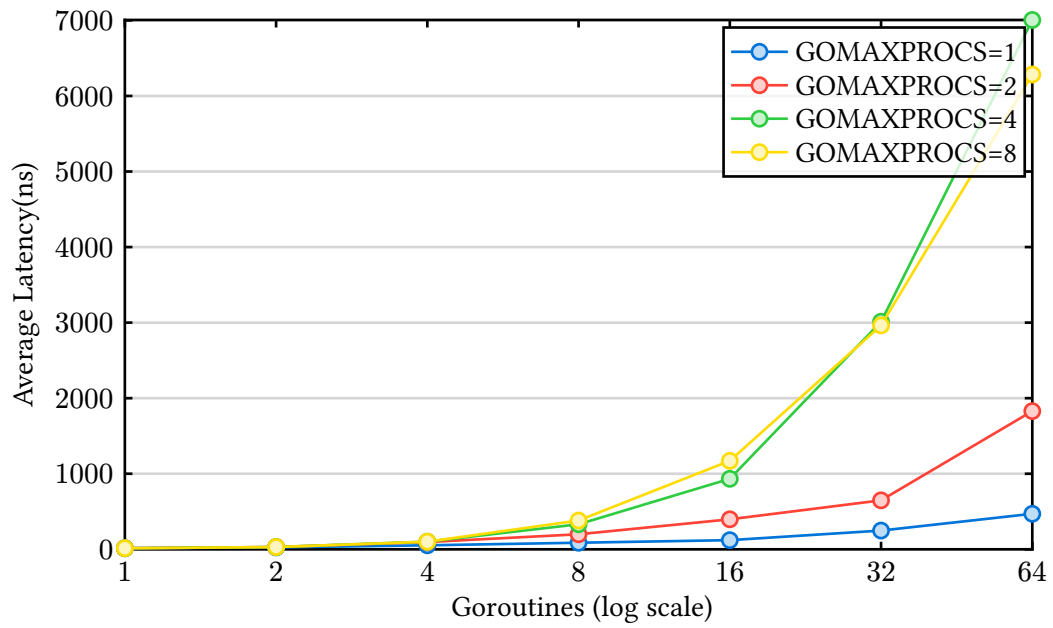


Figure 5: Mutex-bound Throughput Comparison

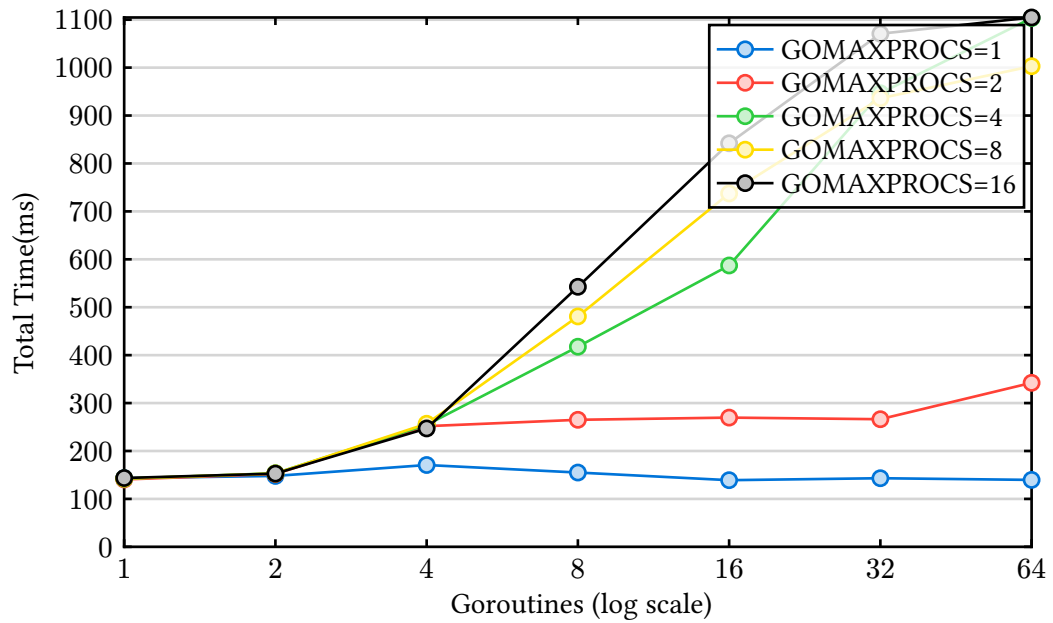


Figure 6: Mutex-bound Throughput Comparison

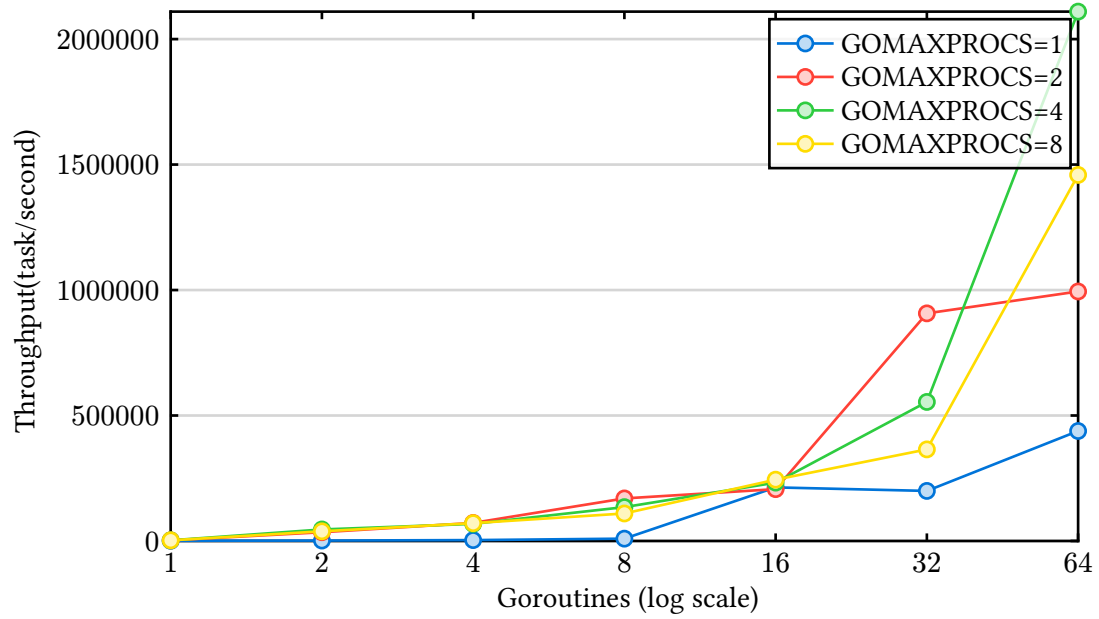


Figure 7: Sleep-bound Throughput Comparison

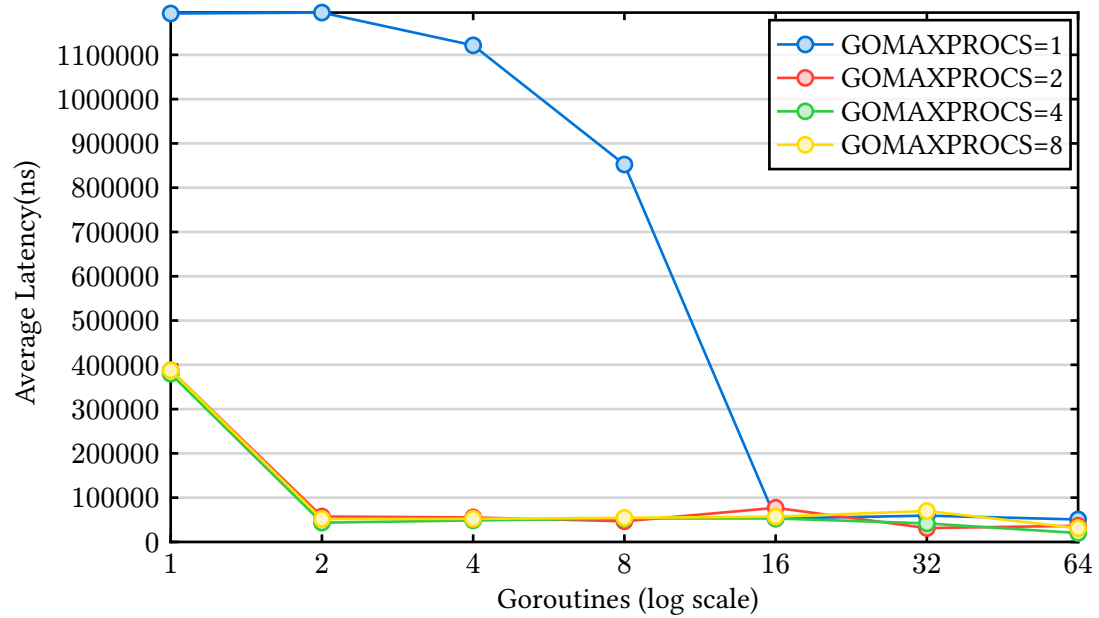


Figure 8: Sleep-bound Throughput Comparison

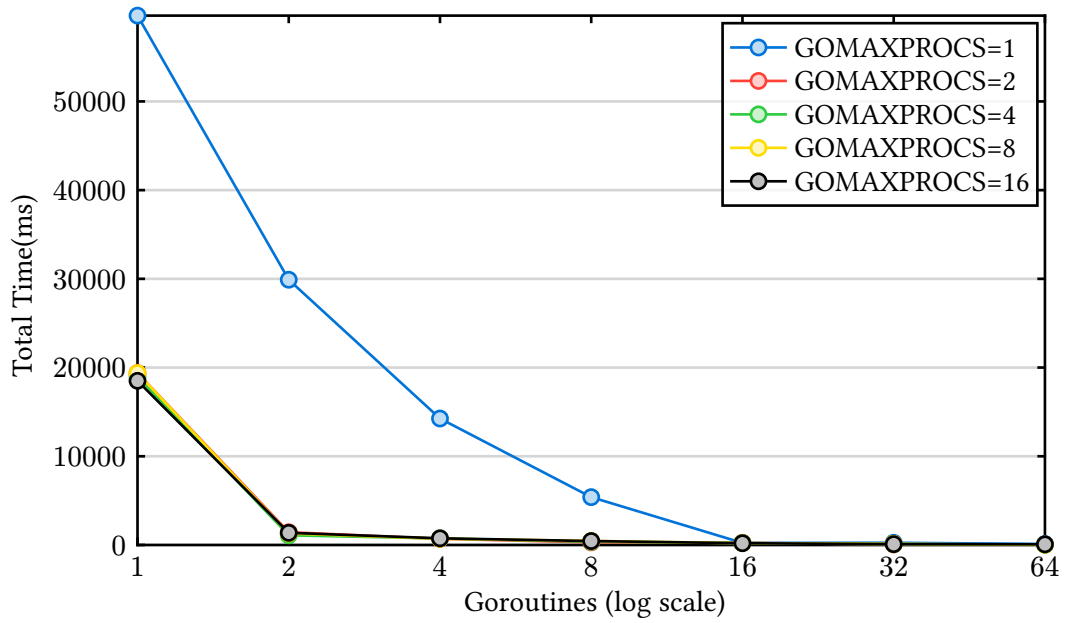


Figure 9: Sleep-bound Throughput Comparison

Workload	GOMAXPROCS	Goroutines	TotalTime_ms	AvgLatency_ns	Throughput
CPU-bound	1	1	402.073	0	1243554167.55
CPU-bound	1	2	402.237	1	1243047962.16
CPU-bound	1	4	403.798	3	1238242153.99
CPU-bound	1	8	398.015	5	1256233338.75
CPU-bound	1	16	401.507	11	1245308834.45
CPU-bound	1	32	403.972	15	1237710034.44
CPU-bound	1	64	407.470	26	1227085470.35
CPU-bound	2	1	401.519	0	1245270717.10
CPU-bound	2	2	200.510	0	2493635904.53
CPU-bound	2	4	199.695	1	2503824378.91
CPU-bound	2	8	208.738	3	2395347013.51
CPU-bound	2	16	202.450	5	2469748141.46
CPU-bound	2	32	202.815	7	2465297766.04
CPU-bound	2	64	200.252	13	2496850136.16
CPU-bound	4	1	404.390	0	1236429188.15
CPU-bound	4	2	203.529	0	2456647589.13
CPU-bound	4	4	104.011	0	4807182006.83
CPU-bound	4	8	119.611	1	4180224772.86
CPU-bound	4	16	113.734	2	4396211362.63
CPU-bound	4	32	106.453	3	4696910039.10
CPU-bound	4	64	107.480	7	4652018805.43
CPU-bound	8	1	404.643	0	1235657800.23
CPU-bound	8	2	202.555	0	2468468316.45
CPU-bound	8	4	104.549	0	4782456846.41

CPU-bound	8	8	92.788	1	5388620115.89
CPU-bound	8	16	88.472	2	5651531751.61
CPU-bound	8	32	78.595	3	6361719103.43
CPU-bound	8	64	75.220	5	6647177054.91
CPU-bound	16	1	405.229	0	1233869616.34
CPU-bound	16	2	202.456	0	2469675362.51
CPU-bound	16	4	115.673	0	4322542819.84
CPU-bound	16	8	94.817	1	5273316800.78
CPU-bound	16	16	60.972	1	8200495824.94
CPU-bound	16	32	103.323	5	4839191910.70
CPU-bound	16	64	62.537	4	7995231516.00
Mutex-bound	1	1	143.326	14	69771157.58
Mutex-bound	1	2	147.955	25	67587943.51
Mutex-bound	1	4	170.733	53	58570829.02
Mutex-bound	1	8	155.018	87	64508588.60
Mutex-bound	1	16	139.075	121	71903548.29
Mutex-bound	1	32	143.170	248	69846997.43
Mutex-bound	1	64	139.672	470	71596380.05
Mutex-bound	2	1	140.586	14	71131055.12
Mutex-bound	2	2	152.138	30	65729800.97
Mutex-bound	2	4	251.772	96	39718546.28
Mutex-bound	2	8	265.074	199	37725245.13
Mutex-bound	2	16	269.675	397	37081719.69
Mutex-bound	2	32	266.294	648	37552498.06
Mutex-bound	2	64	342.247	1829	29218670.03
Mutex-bound	4	1	142.154	14	70346222.94
Mutex-bound	4	2	153.501	30	65146321.77
Mutex-bound	4	4	254.456	101	39299601.09
Mutex-bound	4	8	417.567	331	23948267.87
Mutex-bound	4	16	587.128	933	17032068.72
Mutex-bound	4	32	946.435	3012	10565969.12
Mutex-bound	4	64	1103.286	7006	9063830.25
Mutex-bound	8	1	142.476	14	70187229.07
Mutex-bound	8	2	153.126	30	65305727.10
Mutex-bound	8	4	256.842	101	38934374.58
Mutex-bound	8	8	480.773	381	20799844.72
Mutex-bound	8	16	737.400	1171	13561157.62
Mutex-bound	8	32	936.097	2963	10682657.17
Mutex-bound	8	64	1002.910	6284	9970986.38
Mutex-bound	16	1	143.589	14	69643033.00

Mutex-bound	16	2	152.809	30	65441090.72
Mutex-bound	16	4	247.048	98	40477923.49
Mutex-bound	16	8	542.626	432	18428899.16
Mutex-bound	16	16	842.095	1323	11875138.47
Mutex-bound	16	32	1070.726	3380	9339456.24
Mutex-bound	16	64	1104.659	6923	9052563.87
Sleep-bound	1	1	59671.649	1193432	837.92
Sleep-bound	1	2	29899.159	1195454	1672.29
Sleep-bound	1	4	14254.615	1121510	3507.64
Sleep-bound	1	8	5392.264	852372	9272.54
Sleep-bound	1	16	234.188	52879	213503.47
Sleep-bound	1	32	250.668	59448	199466.85
Sleep-bound	1	64	114.092	50735	438242.55
Sleep-bound	2	1	19384.248	387684	2579.41
Sleep-bound	2	2	1435.552	57389	34829.80
Sleep-bound	2	4	694.912	55350	71951.61
Sleep-bound	2	8	294.829	46603	169589.89
Sleep-bound	2	16	242.050	77073	206568.55
Sleep-bound	2	32	55.124	31345	907053.26
Sleep-bound	2	64	50.288	36501	994282.54
Sleep-bound	4	1	18985.858	379716	2633.54
Sleep-bound	4	2	1089.870	43593	45877.02
Sleep-bound	4	4	733.809	48610	68137.60
Sleep-bound	4	8	370.607	53040	134913.63
Sleep-bound	4	16	215.016	53077	232540.31
Sleep-bound	4	32	90.239	41867	554085.52
Sleep-bound	4	64	23.702	20182	2109507.49
Sleep-bound	8	1	19369.544	387390	2581.37
Sleep-bound	8	2	1297.241	51840	38543.33
Sleep-bound	8	4	704.081	52224	71014.60
Sleep-bound	8	8	456.193	54409	109602.69
Sleep-bound	8	16	204.626	57065	244347.78
Sleep-bound	8	32	136.992	69923	364984.90
Sleep-bound	8	64	34.276	31149	1458751.75
Sleep-bound	16	1	18503.473	370069	2702.20
Sleep-bound	16	2	1371.014	54838	36469.37
Sleep-bound	16	4	750.541	57668	66618.57
Sleep-bound	16	8	444.315	53907	112532.77
Sleep-bound	16	16	198.275	49813	252174.80
Sleep-bound	16	32	94.697	52238	528000.47

Sleep-bound	16	64	73.511	50305	680166.25
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